Class 14: RNASeq Mini Project

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Background

The data for today's mini-project comes from a knock-down study of an important Hox gene

Data Import

```
countFile <- "GSE37704_featurecounts.csv"
metaFile  <- "GSE37704_metadata.csv"

countData <- read.csv(countFile, row.names = 1)
colData   <- read.csv(metaFile,  row.names = 1)
```

Cleanup (data tidying)

```
countData <- countData[, -1]

to.keep <- rowSums(countData) > 10

countData <- countData[to.keep, ]
```

We need to remove the length column from our `countData` to make the columns match the rows in `colData`

```
head(countData)
```

```
                 SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000279457        23        28        29        29        28        46
ENSG00000187634       124       123       205       207       212       258
ENSG00000188976      1637      1831      2383      1226      1326      1504
ENSG00000187961       120       153       180       236       255       357
ENSG00000187583        24        48        65        44        48        64
ENSG00000187642         4         9        16        14        16        16
```

```
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
ncol(countData) == nrow(colData)
```

```
[1] TRUE
```

DESeq Analysis

```r
library(DESeq2)
```

Setting up the input object

```r
dds <- DESeqDataSetFromMatrix(
  countData = countData,
  colData   = colData,
  design    = ~ condition
)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

Running DESeq

```r
dds <- DESeq(dds)
```

```
estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing
```

Getting results

```r
res <- results(dds)
head(res)
```

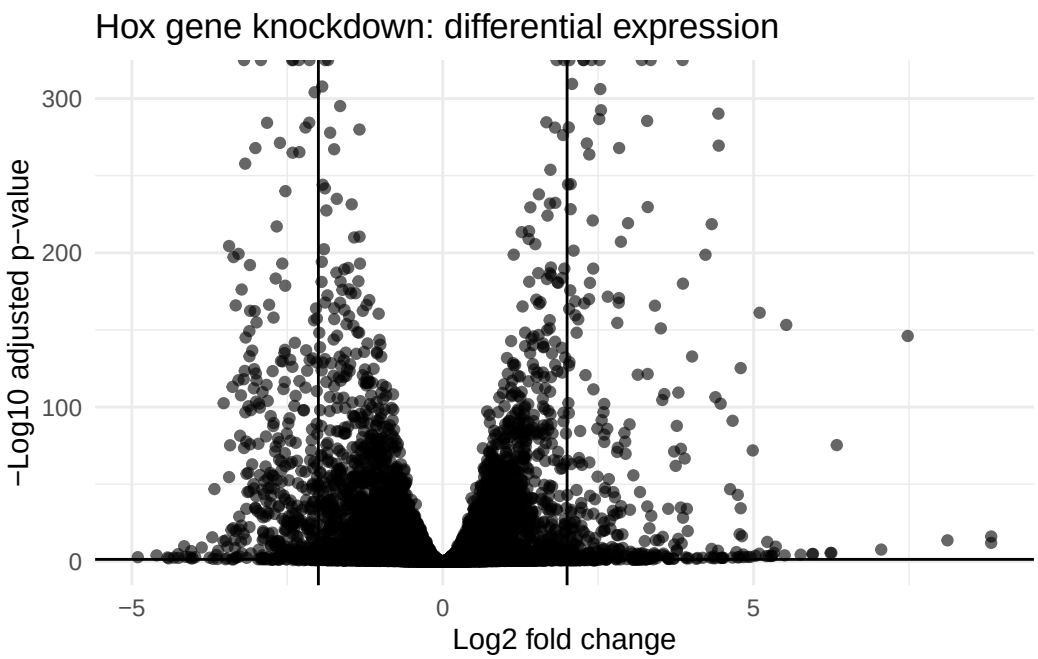
```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 6 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9120      0.1800352 0.3146287   0.572215 5.67176e-01
ENSG00000187634  183.2182      0.4259359 0.1364245   3.122137 1.79543e-03
ENSG00000188976 1651.1025     -0.6927959 0.0548876 -12.622082 1.59547e-36
ENSG00000187961  209.6244      0.7298671 0.1285341   5.678394 1.35965e-08
ENSG00000187583   47.2512      0.0395066 0.2628513   0.150300 8.80528e-01
ENSG00000187642   11.9786      0.5401497 0.5043073   1.071073 2.84137e-01
                      padj
                 <numeric>
ENSG00000279457 6.52317e-01
ENSG00000187634 3.65952e-03
ENSG00000188976 1.79740e-35
ENSG00000187961 4.65764e-08
ENSG00000187583 9.11673e-01
ENSG00000187642 3.67241e-01
```

```r
write.csv(res, "results.csv")
```

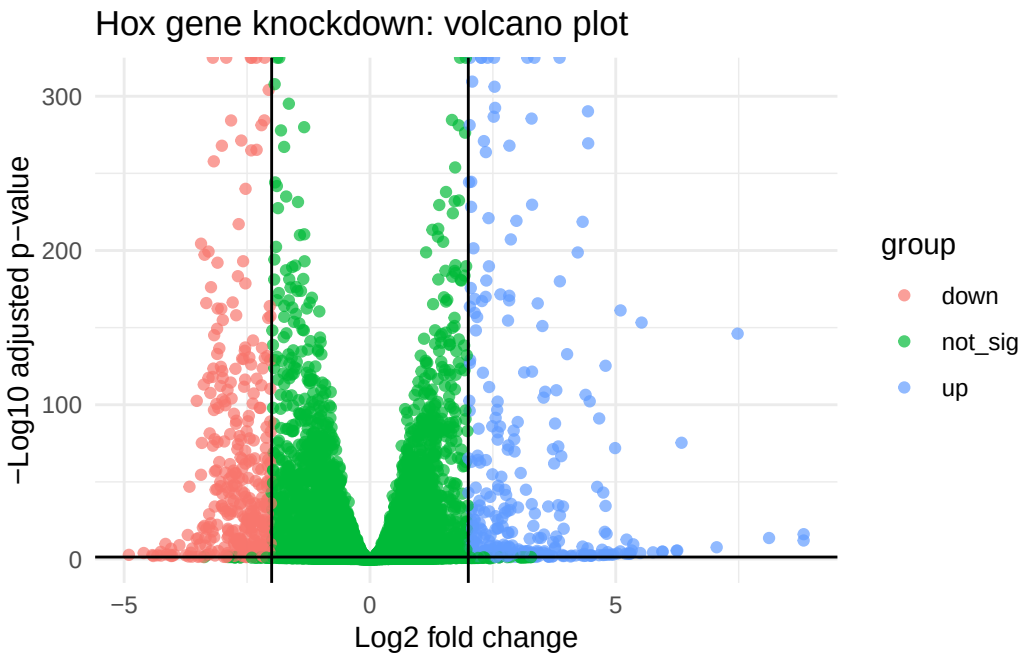
Volcano Plot

```r
library(ggplot2)

res_df <- as.data.frame(res)
res_df <- res_df[!is.na(res_df$padj), , drop = FALSE]

ggplot(res_df, aes(x = log2FoldChange, y = -log10(padj))) +
  geom_point(alpha = 0.6) +
  geom_vline(xintercept = c(-2, 2)) +
  geom_hline(yintercept = -log10(0.05)) +
  labs(
```

```
    title = "Hox gene knockdown: differential expression",
    x = "Log2 fold change",
    y = "-Log10 adjusted p-value"
  ) +
  theme_minimal()
```




```
res_df$group <- "not_sig"
res_df$group[res_df$padj < 0.05 & res_df$log2FoldChange >  2] <- "up"
res_df$group[res_df$padj < 0.05 & res_df$log2FoldChange < -2] <- "down"

ggplot(res_df, aes(x = log2FoldChange, y = -log10(padj), color = group)) +
  geom_point(alpha = 0.7) +
  geom_vline(xintercept = c(-2, 2)) +
  geom_hline(yintercept = -log10(0.05)) +
  labs(
    title = "Hox gene knockdown: volcano plot",
    x = "Log2 fold change",
    y = "-Log10 adjusted p-value"
  ) +
  theme_minimal()
```



Add Annotation (gene symbols and entrez ids)

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys = rownames(res), # ids
                     keytype = "ENSEMBL",# format
                     column = "SYMBOL") # translating to
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$entrez <- mapIds(org.Hs.eg.db,
                     keys = rownames(res), # ids
                     keytype = "ENSEMBL",# format
                     column = "ENTREZID") # translating to
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res$name <- mapIds(org.Hs.eg.db,
                     keys = rownames(res), # ids
                     keytype = "ENSEMBL",# format
                     column = "GENENAME") # translating to
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
res = res[order(res$pvalue),]
write.csv(res, file="deseq_results.csv")

head(res)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 9 columns
                 baseMean log2FoldChange     lfcSE      stat    pvalue
                <numeric>      <numeric> <numeric> <numeric> <numeric>
ENSG00000117519   4483.44       -2.42277 0.0609018  -39.7816         0
ENSG00000183508   2053.71        3.20181 0.0727131   44.0335         0
ENSG00000159176   5692.21       -2.31380 0.0585392  -39.5256         0
ENSG00000116016   4423.77       -1.88809 0.0440078  -42.9034         0
ENSG00000164251   2348.58        3.34442 0.0694188   48.1775         0
ENSG00000124766   2576.46        2.39219 0.0621624   38.4829         0
                     padj      symbol      entrez                   name
                <numeric> <character> <character>            <character>
ENSG00000117519         0        CNN3        1266             calponin 3
ENSG00000183508         0      TENT5C       54855 terminal nucleotidyl..
ENSG00000159176         0       CSRP1        1465 cysteine and glycine..
ENSG00000116016         0       EPAS1        2034 endothelial PAS doma..
ENSG00000164251         0       F2RL1        2150 F2R like trypsin rec..
ENSG00000124766         0        SOX4        6659 SRY-box transcriptio..
```

Pathway Analysis

```r
library(gage)
library(gageData)
library(pathview)
```

```r
foldchanges <- res$log2FoldChange
names(foldchanges) <- res$symbol
head(foldchanges)
```

```
     CNN3    TENT5C     CSRP1     EPAS1     F2RL1      SOX4
-2.422773  3.201809 -2.313796 -1.888087  3.344422  2.392190
```

KEGG

```r
data(kegg.sets.hs)

keggres <- gage(foldchanges, gsets = kegg.sets.hs)
head(keggres$less, 5)
```

```
                                        p.geomean stat.mean p.val q.val
hsa00232 Caffeine metabolism                   NA       NaN    NA    NA
hsa00983 Drug metabolism - other enzymes       NA       NaN    NA    NA
hsa01100 Metabolic pathways                    NA       NaN    NA    NA
hsa00230 Purine metabolism                     NA       NaN    NA    NA
hsa05340 Primary immunodeficiency              NA       NaN    NA    NA
                                        set.size exp1
hsa00232 Caffeine metabolism                    0   NA
hsa00983 Drug metabolism - other enzymes        0   NA
hsa01100 Metabolic pathways                     0   NA
hsa00230 Purine metabolism                      0   NA
hsa05340 Primary immunodeficiency               0   NA
```

```r
pathview(gene.data = foldchanges, pathway.id = "hsa04110")
```

```
Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/matthewhuang/BIMM 143 correct/class14

Info: Writing image file hsa04110.pathview.png
```

```r
pathview(gene.data = foldchanges, pathway.id = "hsa04114")
```

```
Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/matthewhuang/BIMM 143 correct/class14

Info: Writing image file hsa04114.pathview.png
```

```r
pathview(gene.data = foldchanges, pathway.id = "hsa03030")
```

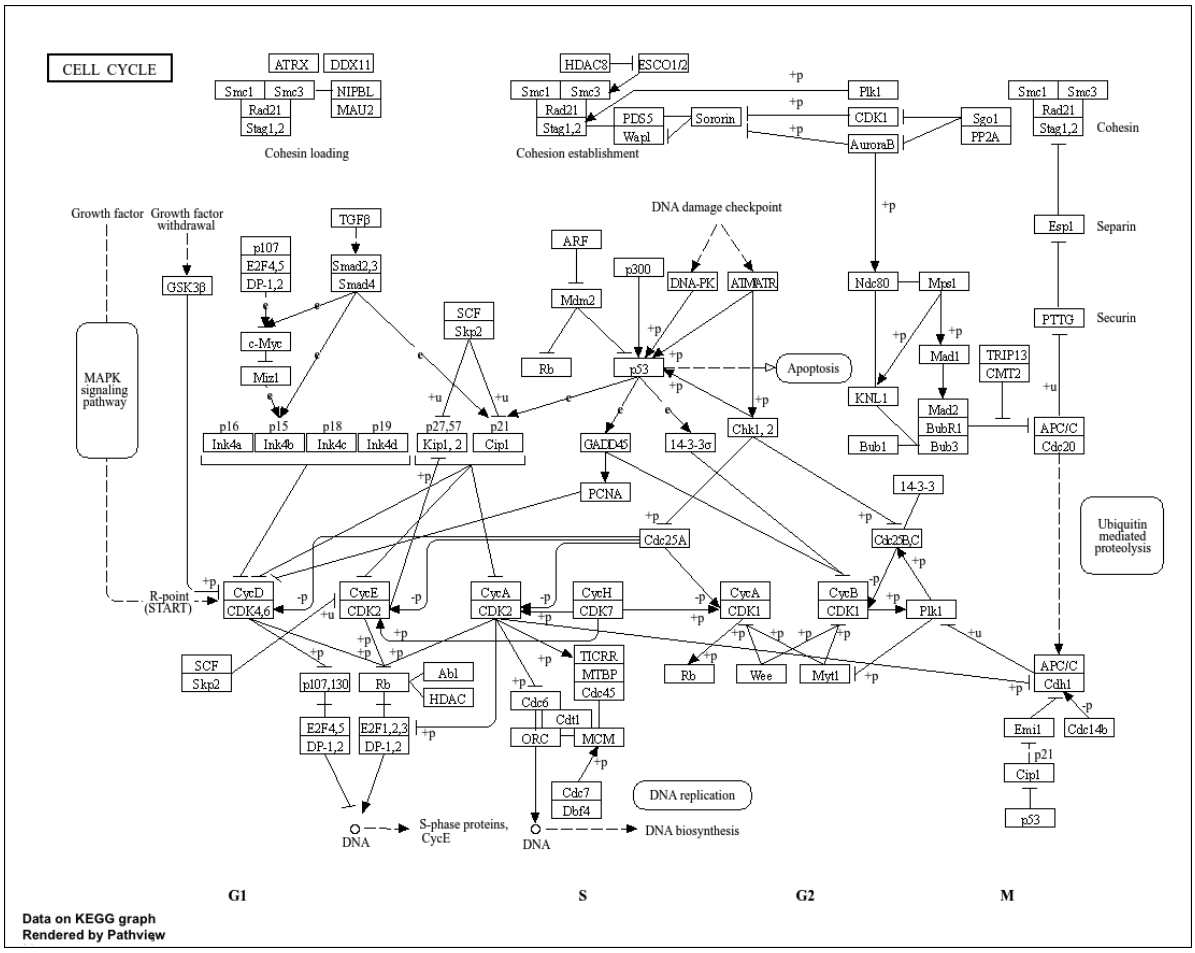
```
Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/matthewhuang/BIMM 143 correct/class14

Info: Writing image file hsa03030.pathview.png
```

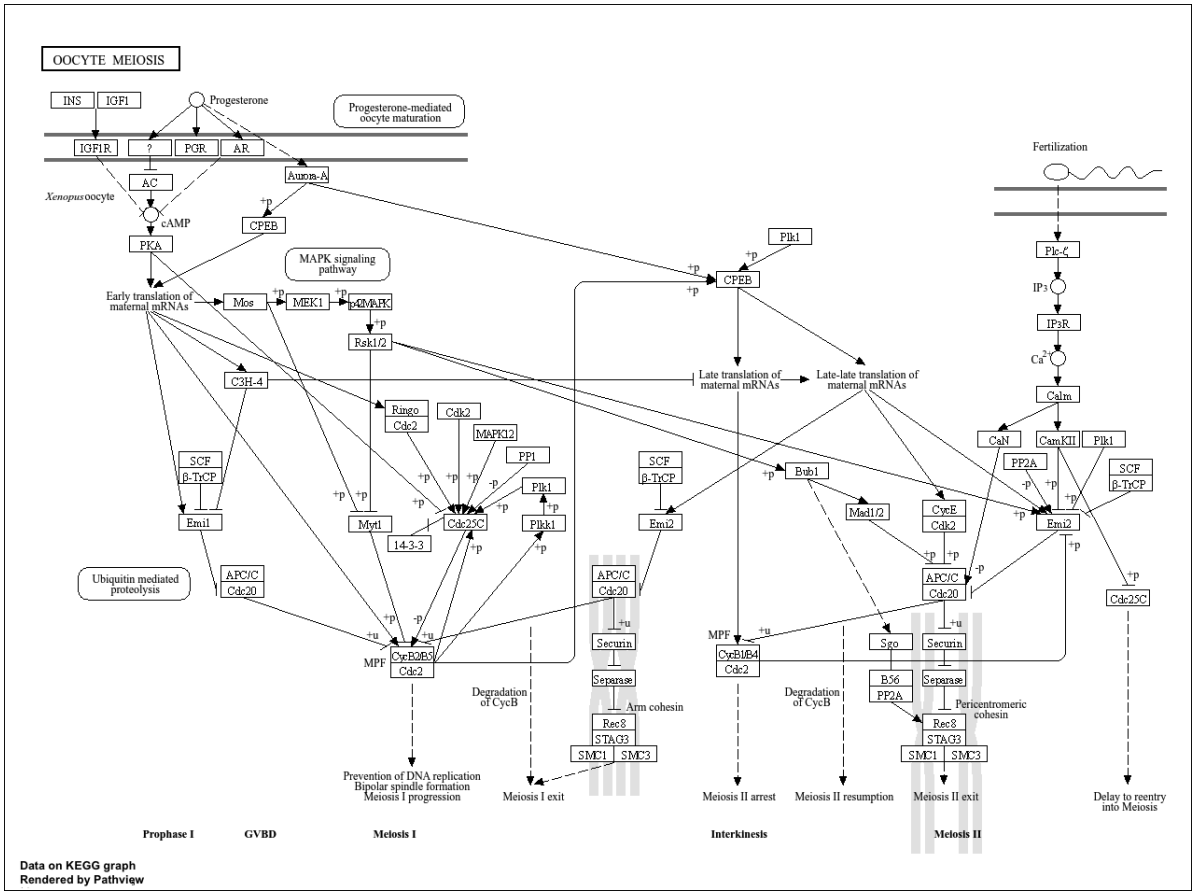
```r
pathview(gene.data = foldchanges, pathway.id = "hsa05130")
```

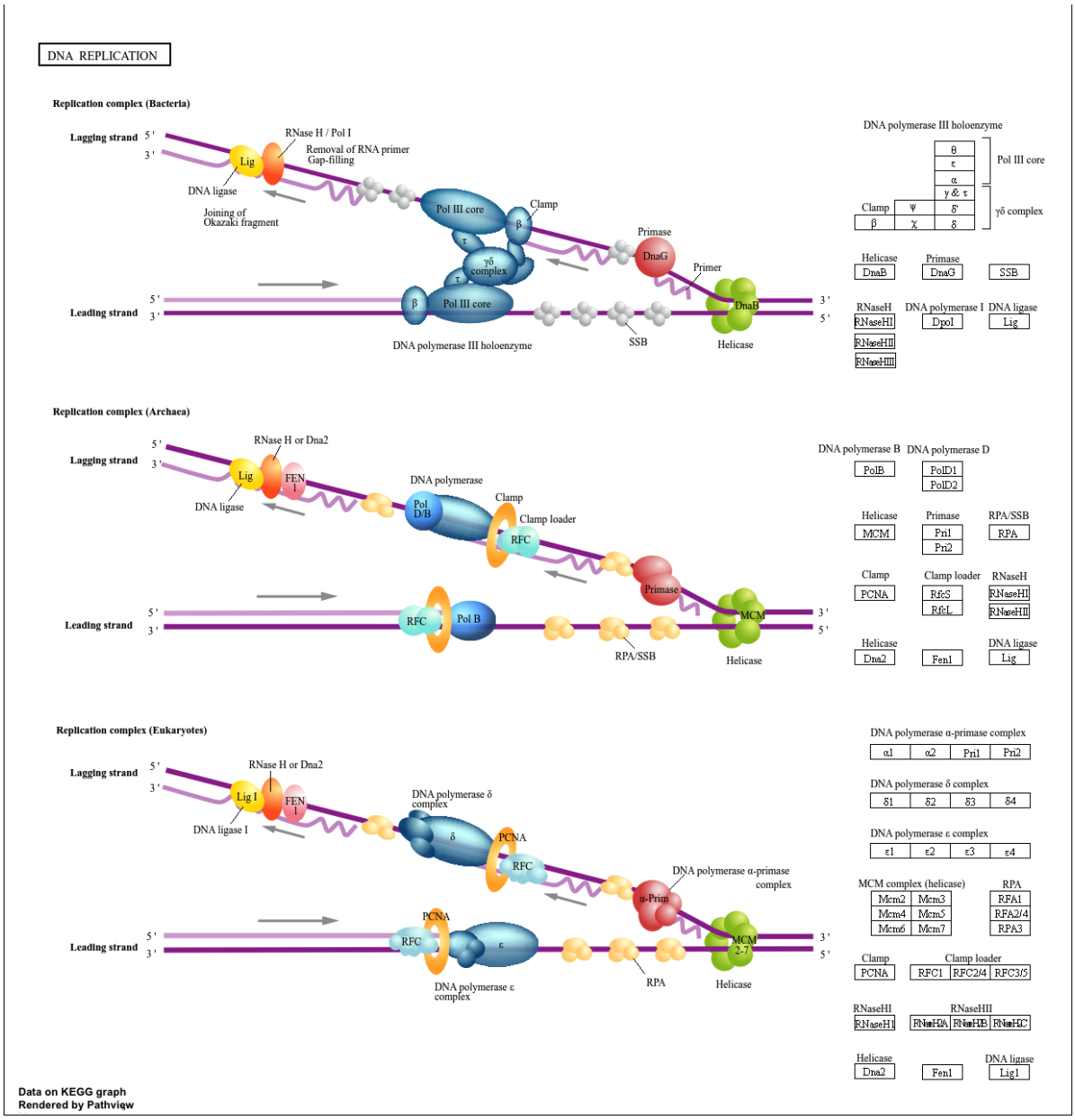
```
Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/matthewhuang/BIMM 143 correct/class14

Info: Writing image file hsa05130.pathview.png
```

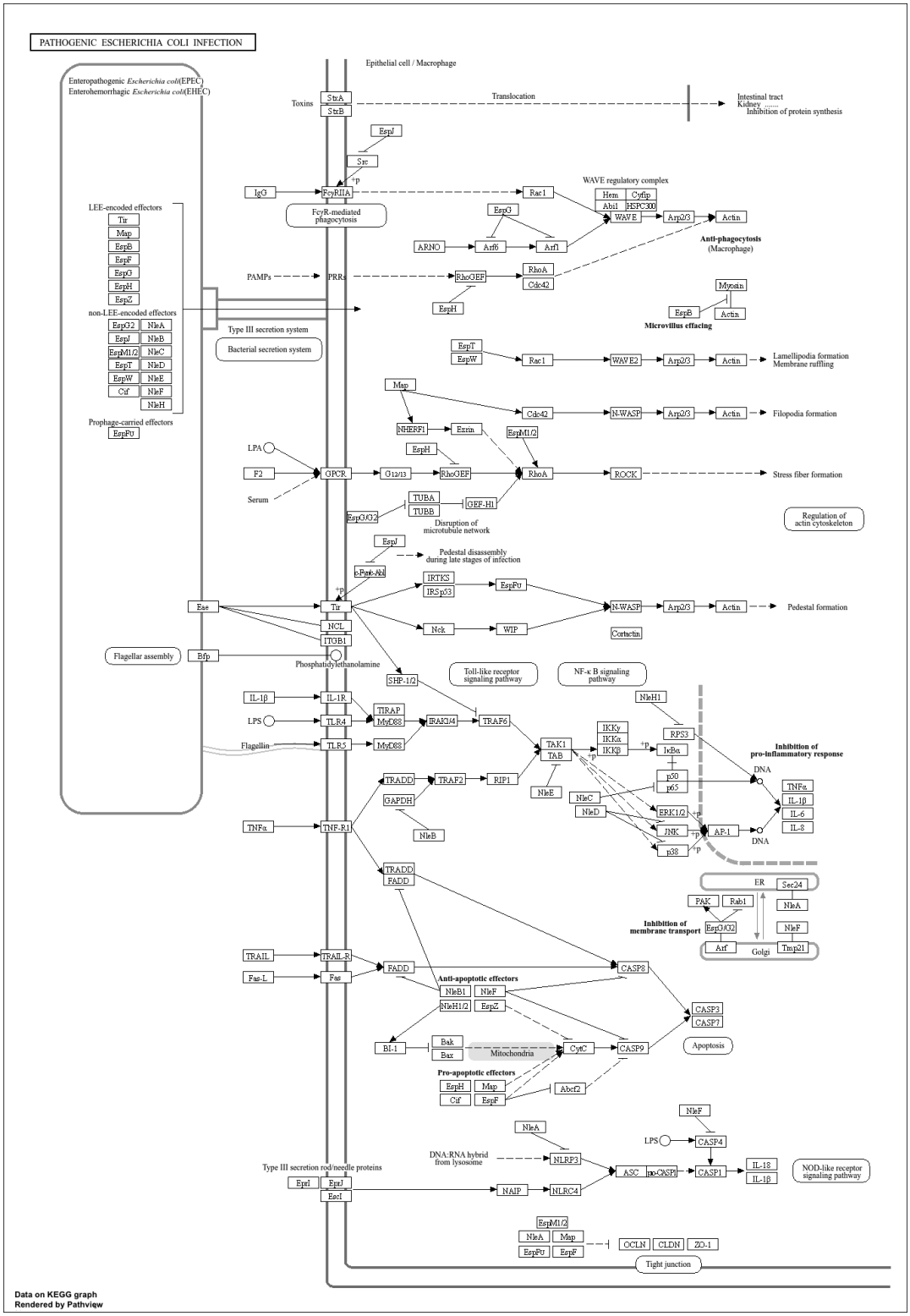
```r
pathview(gene.data = foldchanges, pathway.id = "hsa03440")
```

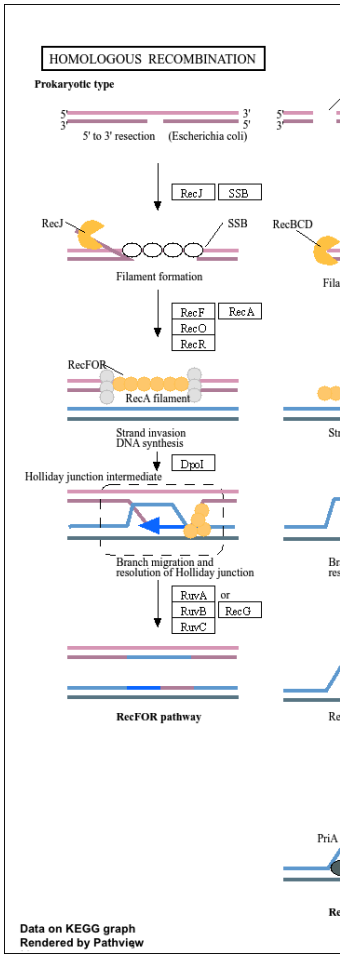
```
Warning: None of the genes or compounds mapped to the pathway!
Argument gene.idtype or cpd.idtype may be wrong.

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/matthewhuang/BIMM 143 correct/class14

Info: Writing image file hsa03440.pathview.png
```




OOCYTE MEIOSIS
INS
IGF1
Progesterone
Progesterone-mediated oocyte maturation
IGF1R
?
PGR
AR
AC
Xenopus oocyte
cAMP
PKA
Aurora-A
CPEB
MAPK signaling pathway
Early translation of maternal mRNAs
Mos
MEK1
p42MAPK
Rsk1/2
C3H-4
Ringo
Cdc2
Cdk2
MAPK12
PP1
Plk1
SCF
β-TrCP
Emi1
Myt1
14-3-3
Cdc25C
Plkk1
Ubiquitin mediated proteolysis
APC/C
Cdc20
CycB2/B5
MPF
Degradation of CycB
Prevention of DNA replication
Bipolar spindle formation
Meiosis I progression
Meiosis I exit
Securin
Separase
Arm cohesin
Rec8
STAG3
SMC1
SMC3
Emi2
Late translation of maternal mRNAs
Late-late translation of maternal mRNAs
Bub1
Mad1/2
CycE
Cdk2
CycB1/B4
Sgo
B56
PP2A
Pericentromeric cohesin
Meiosis II arrest
Meiosis II resumption
Meiosis II exit
Fertilization
Plc-ζ
IP3
IP3R
Ca2+
Calm
CaN
CamKII
Delay to reentry into Meiosis
Prophase I
GVBD
Meiosis I
Interkinesis
Meiosis II
Data on KEGG graph
Rendered by Pathview


DNA REPLICATION
Replication complex (Bacteria)
Lagging strand
5'
3'
RNase H / Pol I
Removal of RNA primer
Gap-filling
Lig
DNA ligase
Joining of
Okazaki fragment
Pol III core
Clamp
β
τ
γδ
complex
Primase
DnaG
Primer
Leading strand
DNA polymerase III holoenzyme
SSB
DnaB
Helicase
DNA polymerase III holoenzyme
θ
ε
α
γ & τ
γδ complex
Pol III core
Clamp
ψ
δ'
β
χ
δ
Helicase
DnaB
Primase
DnaG
SSB
RNaseH
RNaseHI
RNaseHII
RNaseHIII
DNA polymerase I
DpoI
DNA ligase
Lig
Replication complex (Archaea)
RNase H or Dna2
FEN
1
DNA polymerase
Pol
D/B
Clamp loader
RFC
Primase
MCM
RPA/SSB
DNA polymerase B
PolB
DNA polymerase D
PolD1
PolD2
Helicase
MCM
Primase
Pri1
Pri2
RPA/SSB
RPA
Clamp
PCNA
Clamp loader
RfcS
RfcL
RNaseH
RNaseHI
RNaseHII
Helicase
Dna2
Fen1
DNA ligase
Lig
Replication complex (Eukaryotes)
Lig I
DNA ligase I
DNA polymerase δ
complex
δ
PCNA
DNA polymerase α-primase
complex
α-Prim
ε
DNA polymerase ε
complex
MCM
2-7
DNA polymerase α-primase complex
α1
α2
Pri1
Pri2
DNA polymerase δ complex
δ1
δ2
δ3
δ4
DNA polymerase ε complex
ε1
ε2
ε3
ε4
MCM complex (helicase)
Mcm2
Mcm3
Mcm4
Mcm5
Mcm6
Mcm7
RPA
RFA1
RFA2/4
RFA3
Clamp
PCNA
Clamp loader
RFC1
RFC2/4
RFC3/5
RNaseHI
RNaseHI1
RNaseHIII
RNaseHDA
RNaseHDB
RNaseHDC
Helicase
Dna2
Fen1
DNA ligase
Lig1
Data on KEGG graph
Rendered by Pathview


PATHOGENIC ESCHERICHIA COLI INFECTION
Epithelial cell / Macrophage
Enteropathogenic Escherichia coli(EPEC)
Enterohemorrhagic Escherichia coli(EHEC)
Toxins
StxA
StxB
Translocation
Intestinal tract
Kidney
Inhibition of protein synthesis
EspJ
Src
IgG
FcγRIIA
FcγR-mediated phagocytosis
Rac1
WAVE regulatory complex
Hem
Cyfip
Abi1
HSPC300
WAVE
Arp2/3
Actin
EspG
ARNO
Arf6
Arf1
Anti-phagocytosis
(Macrophage)
LEE-encoded effectors
Tir
Map
EspB
EspF
EspG
EspH
EspZ
PAMPs
PRRs
RhoGEF
RhoA
Cdc42
EspH
Myosin
EspB
Actin
Microvillus effacing
non-LEE-encoded effectors
EspG2
NleA
EspJ
NleB
EspM1/2
NleC
EspT
NleD
EspW
NleE
Cif
NleF
NleH
Type III secretion system
Bacterial secretion system
EspT
EspW
Rac1
WAVE2
Arp2/3
Actin
Lamellipodia formation
Membrane ruffling
Prophage-carried effectors
EspFu
Map
Cdc42
N-WASP
Arp2/3
Actin
Filopodia formation
NHERF1
Ezrin
EspM1/2
EspH
LPA
F2
GPCR
G12/13
RhoGEF
RhoA
ROCK
Stress fiber formation
Serum
TUBA
TUBB
GEF-H1
EspG/G2
Disruption of
microtubule network
Regulation of
actin cytoskeleton
EspJ
Pedestal disassembly
during late stages of infection
c-Src/c-Abl
IRTKS
IRSp53
EspFu
Eae
Tir
NCL
ITGB1
N-WASP
Arp2/3
Actin
Pedestal formation
Nck
WIP
Cortactin
Flagellar assembly
Bfp
Phosphatidylethanolamine
SHP-1/2
Toll-like receptor
signaling pathway
NF-κ B signaling
pathway
NleH1
IL-1β
IL-1R
LPS
TLR4
TIRAP
MyD88
IRAK1/4
TRAF6
Flagellin
TLR5
MyD88
TAK1
TAB
IKKγ
IKKα
IKKβ
RPS3
IκBα
Inhibition of
pro-inflammatory response
DNA
p50
p65
TNFα
IL-1β
IL-6
IL-8
TRADD
TRAF2
RIP1
GAPDH
NleE
NleC
NleD
ERK1/2
JNK
p38
AP-1
DNA
NleB
TNFα
TNF-R1
TRADD
FADD
ER
Sec24
PAK
Rab1
NleA
Inhibition of
membrane transport
EspG/G2
NleF
Arf
Golgi
Trap21
TRAIL
TRAIL-R
Fas-L
Fas
FADD
CASP8
Anti-apoptotic effectors
NleB1
NleF
NleH1/2
EspZ
CASP3
CASP7
BI-1
Bak
Bax
Mitochondria
CytC
CASP9
Apoptosis
Pro-apoptotic effectors
EspH
Map
Cif
EspF
Abcf2
NleF
NleA
LPS
CASP4
DNA:RNA hybrid
from lysosome
NLRP3
ASC
pro-CASP1
CASP1
IL-18
IL-1β
NOD-like receptor
signaling pathway
Type III secretion rod/needle proteins
EprI
EprJ
EscI
NAIP
NLRC4
EspM1/2
NleA
Map
EspFu
EspF
OCLN
CLDN
ZO-1
Tight junction
Data on KEGG graph
Rendered by Pathview


HOMOLOGOUS RECOMBINATION
Prokaryotic type
5' to 3' resection
(Escherichia coli)
RecJ
SSB
RecJ
SSB
RecBCD
Filament formation
RecF
RecA
RecO
RecR
RecFOR
RecA filament
Strand invasion
DNA synthesis
DpoI
Holliday junction intermediate
Branch migration and
resolution of Holliday junction
RuvA
or
RuvB
RecG
RuvC
RecFOR pathway
PriA
Data on KEGG graph
Rendered by Pathview

GO

```
data(go.sets.hs)
data(go.subs.hs)

gobpsets <- go.sets.hs[go.subs.hs$BP]
gobpres  <- gage(foldchanges, gsets = gobpsets)

head(gobpres$less)
```

```
                                            p.geomean stat.mean p.val q.val
GO:0000002 mitochondrial genome maintenance        NA       NaN    NA    NA
GO:0000003 reproduction                            NA       NaN    NA    NA
GO:0000012 single strand break repair              NA       NaN    NA    NA
GO:0000018 regulation of DNA recombination         NA       NaN    NA    NA
GO:0000019 regulation of mitotic recombination     NA       NaN    NA    NA
GO:0000022 mitotic spindle elongation              NA       NaN    NA    NA
                                            set.size exp1
GO:0000002 mitochondrial genome maintenance        0   NA
GO:0000003 reproduction                            0   NA
GO:0000012 single strand break repair              0   NA
GO:0000018 regulation of DNA recombination         0   NA
GO:0000019 regulation of mitotic recombination     0   NA
GO:0000022 mitotic spindle elongation              0   NA
```

Reactome

```
sig_genes <- res[!is.na(res$padj) & res$padj <= 0.05, "symbol"]
sig_genes <- sig_genes[!is.na(sig_genes)]

cat("Total number of significant genes:", length(sig_genes), "\n")
```

```
Total number of significant genes: 8262
```

```
write.table(
  sig_genes,
  file = "significant_genes.txt",
  row.names = FALSE,
```

```
  col.names = FALSE,
  quote = FALSE
)
```

> Q: What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

Cell Cycle (Mitotic) was the most significant pathway. Overall, the most significant pathways largely align with the earlier KEGG results. Differences between the two methods likely come from how each database organizes and defines pathways. KEGG tends to group processes more broadly, like listing the cell cycle as a single pathway, while Reactome breaks the same biological processes into multiple, more specific and detailed pathways.